

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

Paper 2

Tuesday 30 May 2023

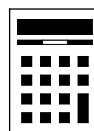
07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	



J U N 2 3 9 2 0 2 2 0 1

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Answer **all** questions in the spaces provided.**0 1**

This question is about molecules.

0 1 . 1

What type of bond holds the atoms together in a molecule which is formed from non-metals?

[1 mark]Tick (✓) **one** box.

Covalent

☐

Ionic

☐

Metallic

☐**0 1 . 2**Ammonia (NH₃) is a simple molecule.

How many bonds does an atom of nitrogen form in a molecule of ammonia?

[1 mark]Tick (✓) **one** box.

1

☐

2

☐

3

☐

4

☐

0 1 . 3 Table 1 shows some properties of four substances: **A**, **B**, **C** and **D**.

Table 1

Substance	Melting point in °C	Electrical conductivity when solid
A	−39	Good
B	769	Good
C	−66	Poor
D	1723	Poor

Which substance consists of simple molecules?

[1 mark]

Tick (✓) **one** box.

A ☐ **B** ☐ **C** ☐ **D** ☐

0 1 . 4 Crude oil is a mixture of hydrocarbon molecules.

Complete the sentence.

Choose the answer from the box.

[1 mark]

chromatography	crystallisation	distillation	filtration
----------------	-----------------	--------------	------------

Hydrocarbon molecules are separated from crude oil by

_____.

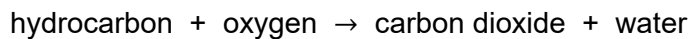
Question 1 continues on the next page

Turn over ►



0 1 . 5 The combustion of a hydrocarbon produces molecules of carbon dioxide and water.

The word equation for the reaction is:



What happens to the hydrogen in the hydrocarbon during combustion?

[1 mark]

Tick (✓) **one** box.

The hydrogen dissolves

☐

The hydrogen neutralises

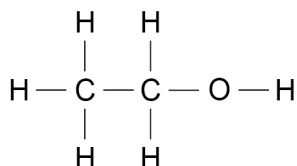
☐

The hydrogen oxidises

☐

0 1 . 6 **Figure 1** shows the displayed structure of a molecule of ethanol.

Figure 1



Explain why ethanol is **not** a hydrocarbon.

[2 marks]

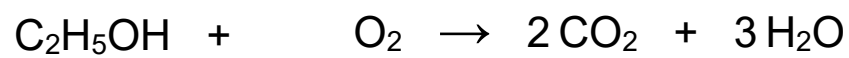


01.7

Ethanol combusts in oxygen.

Balance the equation for the reaction.

[1 mark]

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Turn over for the next question

Turn over ►



0 2

This question is about atoms.

0 2 . 1

Complete **Table 2** to show the charge and relative mass of each particle in an atom.**[3 marks]****Table 2**

Name of Particle	Charge	Relative mass
Electron		Very small
Neutron		1
Proton	+1	

0 2 . 2

Why does an atom have no overall charge?

[1 mark]

0 2 . 3

An atom of an isotope of chlorine is represented as $^{35}_{17}\text{Cl}$ A different isotope of chlorine has two more neutrons than $^{35}_{17}\text{Cl}$ Which of the following represents an atom with two more neutrons than $^{35}_{17}\text{Cl}$?**[1 mark]**Tick (✓) **one** box. $^{35}_{17}\text{Cl}$ ☐ $^{35}_{19}\text{Cl}$ ☐ $^{37}_{17}\text{Cl}$ ☐ $^{37}_{19}\text{Cl}$ ☐

0 2 . 4 What is the relative atomic mass (A_r) of an element?

[1 mark]

Tick (✓) **one** box.

Average mass of the isotopes of the element compared to ^{12}C

☐

Maximum mass of an isotope of the element compared to ^{12}C

☐

Minimum mass of an isotope of the element compared to ^{12}C

☐

Most common mass of the isotopes of the element compared to ^{12}C

☐

Neon gas contains neon atoms.

0 2 . 5 Calculate the mass of 4 moles of neon atoms.

Relative atomic mass (A_r): Ne = 20

[2 marks]

Mass = _____ g

0 2 . 6 Why do neon atoms **not** form neon compounds?

[1 mark]

Turn over ►



0 3

This question is about elements and compounds.

0 3 . 1

What is meant by a 'compound'?

[1 mark]

0 3 . 2

Which group of elements would form ions with a +1 charge?

[1 mark]Tick (✓) **one** box.

Alkali metals

☐

Halogens

☐

Noble gases

☐**0 3 . 3**

Which element has atoms with the same electron arrangement as chloride ions?

Use the periodic table.

[1 mark]Tick (✓) **one** box.

Argon

☐

Bromine

☐

Fluorine

☐

Sulfur

☐

Sodium chloride is an ionic compound.

0 3 . 4

Solid sodium chloride does **not** conduct electricity.

Give **one** reason why solid sodium chloride does **not** conduct electricity.

[1 mark]

0 3 . 5

The structure of solid sodium chloride is a giant ionic lattice.

Describe the forces of attraction between the ions in solid sodium chloride.

[3 marks]

0 3 . 6

Aluminium nitrate is a compound that contains the ions:

- Al^{3+}
- NO_3^-

Give the formula of aluminium nitrate.

[1 mark]

—
8

Turn over ►



0 4

A teacher investigated the reactivity of four metals: barium, copper, iron and magnesium.

This is the method used.

- 1 Add 10 cm³ of dilute hydrochloric acid to a test tube.
- 2 Add a piece of barium to the hydrochloric acid.
- 3 Observe the rate of bubbling.
- 4 Repeat steps 1 to 3 with copper, iron and magnesium.

0 4 . 1

Name a piece of equipment that could be used to measure 10 cm³ of dilute hydrochloric acid.

[1 mark]

0 4 . 2

Table 3 shows the results.

Table 3

Metal	Reaction with dilute hydrochloric acid
Barium	Bubbles very rapidly
Copper	No reaction
Iron	Bubbles slowly
Magnesium	Bubbles rapidly

Write down the metals in order of reactivity.

Use **Table 3**.

[1 mark]

_____ (Most reactive)

_____ (Least reactive)

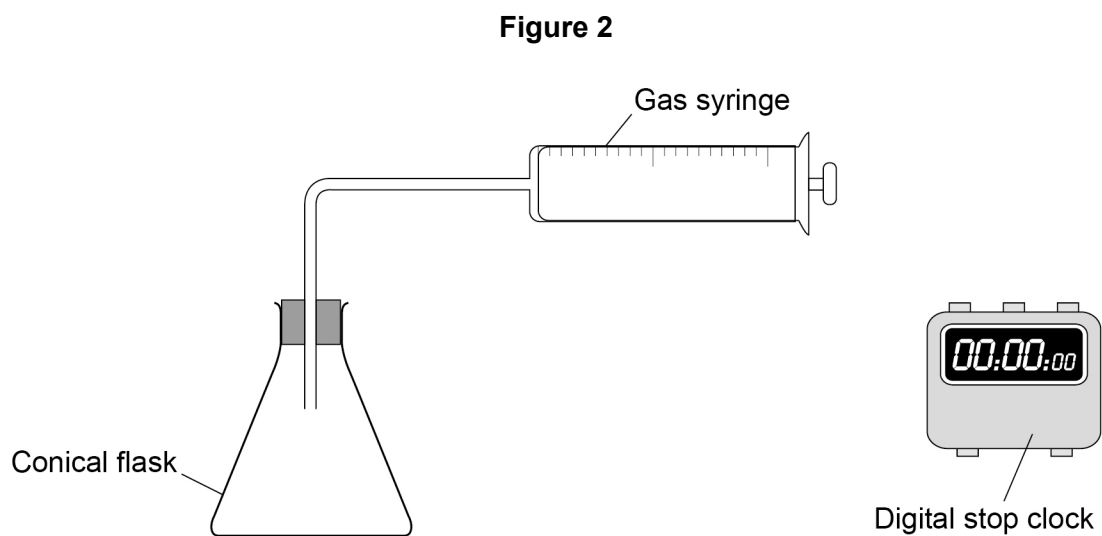


0 4 . 3 What is the independent variable in this investigation?

[1 mark]

0 4 . 4 The teacher repeated the investigation using different apparatus.

Figure 2 shows the apparatus.



Describe how the apparatus could be used to obtain more accurate results for the investigation.

[3 marks]

Question 4 continues on the next page

Turn over ►



The teacher compared the reaction of barium with dilute hydrochloric acid and the reaction of barium with dilute sulfuric acid.

Table 4 shows the results.

Table 4

Reaction with dilute hydrochloric acid	Reaction with dilute sulfuric acid
Bubbles very rapidly	Bubbles at the start The reaction stops after a few seconds Barium is coated in a white solid

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0 4 . 5

Explain why a white solid is produced when barium reacts with dilute sulfuric acid.

[2 marks]

0 4 . 6

The white solid affects the speed of the reaction.

Suggest why the reaction stops after a few seconds.

You should refer to reacting particles.

[1 mark]

9



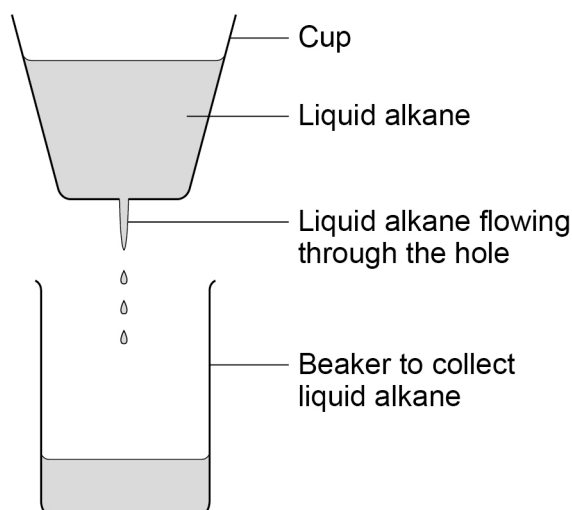
0 5

A student investigated the effect of temperature on the viscosity of a liquid alkane.

The student measured the rate of flow of the alkane through a hole in a cup.

Figure 3 shows the apparatus used to measure the rate of flow.

Figure 3



This is the method used.

- 1 Change the temperature of the alkane to 25 °C.
- 2 Pour 50 cm³ of the alkane into the cup.
- 3 Record the time taken for all of the alkane to flow through the hole.
- 4 Repeat steps 2 to 3 two more times.
- 5 Repeat steps 1 to 4 at different temperatures.

0 5 . 1

The student used a water bath instead of a Bunsen burner to change the temperature of the alkane.

Suggest **one** reason why.

[1 mark]

0 5 . 2

The size of the hole in the cup was kept the same for each experiment.

How does the size of the hole affect the rate of flow of the liquid alkane?

[1 mark]

Question 5 continues on the next page

Turn over ►



Table 5 shows the results.

Table 5

Temperature of alkane in °C	Time taken for alkane to flow through the hole in seconds			Mean time in seconds
	Trial 1	Trial 2	Trial 3	
25	65	66	67	66
30	60	X	55	59
35	54	56	52	54
40	50	52	51	51
45	49	51	47	49
50	47	48	46	47

0 5 . 3 Calculate value **X** in **Table 5**.

[2 marks]

X = _____ seconds

0 5 . 4 Describe the relationship between the temperature of the alkane and the **viscosity** of the alkane.

Use **Table 5**.

[3 marks]



0	5	.	5
---	---	---	---

At a temperature of 55 °C, the alkane took 53 seconds to flow through the cup.

The student did **not** include this result in the table because the result was anomalous.

The student had **not** measured the volume of alkane correctly.

Explain how an incorrectly measured volume caused this anomalous result.

[2 marks]

0	5	.	6
---	---	---	---

The student investigated the viscosity of a different hydrocarbon.

The empirical formula of the hydrocarbon was CH₂

0.31 moles of the hydrocarbon had a mass of 43.4 g.

Determine the molecular formula of the hydrocarbon.

[4 marks]

Molecular formula = _____

13

Turn over ►



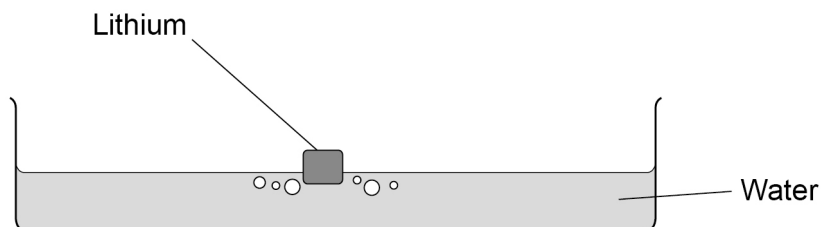
0 6

This question is about Group 1 elements and Group 7 elements.

Group 1 elements react with water.

Figure 4 shows lithium reacting with water.

Figure 4



The following observations are made:

- lithium slowly moves on the surface of the water
- bubbles of gas are slowly produced.

0 6**. 1**

Suggest how the observations would be different if sodium is added to water.

[2 marks]

0 6**. 2**

Which gas is produced when lithium reacts with water?

[1 mark]



0 6 . 3 The lithium reacted with the water to produce an alkaline solution.

Universal indicator was added to the alkaline solution.

What colour did the universal indicator turn?

[1 mark]

0 6 . 4 Metals react with Group 7 elements.

The reaction between iron and bromine forms a solid compound.

Describe a chemical test to identify if the solid compound is iron(II) bromide or iron(III) bromide.

Give the results of the tests with iron(II) bromide and iron(III) bromide.

[4 marks]

Test _____

Result with iron(II) bromide _____

Result with iron(III) bromide _____

8

Turn over for the next question

Turn over ►

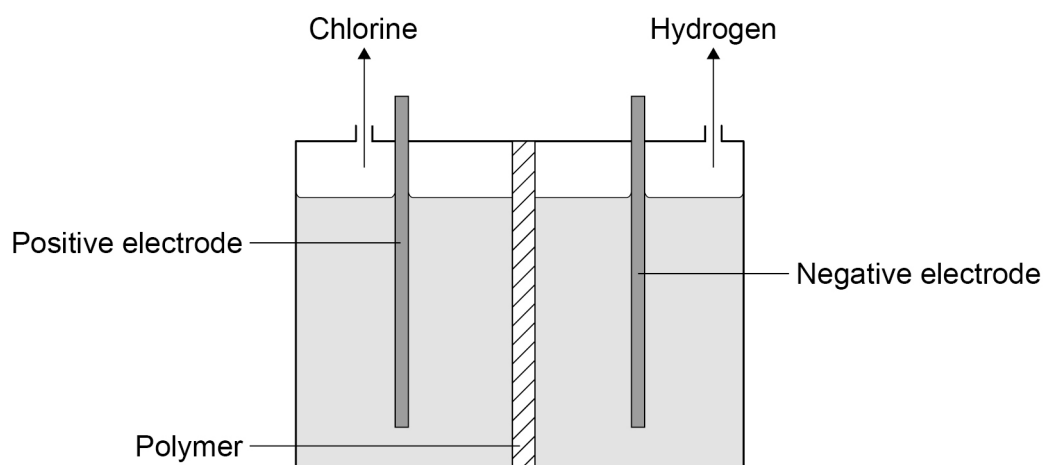


0 7

This question is about electrolysis.

Figure 5 shows the industrial electrolysis of sodium chloride solution.

Figure 5



The polymer divides the electrolysis cell and allows the flow of ions.

0 7**1**

Damp blue litmus paper is used to test for chlorine gas.

Give the results of this test if chlorine is present.

[2 marks]

0 7 . 2 The positive electrode in **Figure 5** is made from the metal titanium.

Suggest **two** reasons why titanium is used.

[2 marks]

1 _____

2 _____

0 7 . 3 Give the reason why hydrogen is formed at the negative electrode instead of sodium.

[1 mark]

0 7 . 4 Complete the half equation for the formation of hydrogen at the negative electrode.

[2 marks]



0 7 . 5 A solution of compound **X** is produced during the electrolysis of sodium chloride solution.

What is the formula of compound **X**?

[1 mark]

8

Turn over ►



0 8

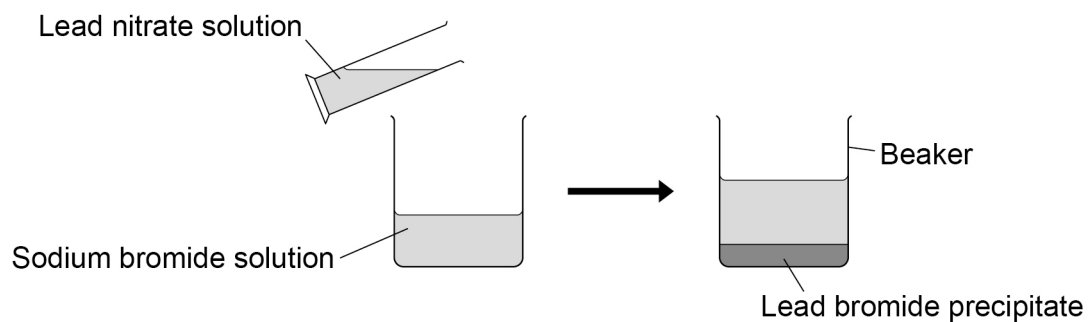
This question is about insoluble salts.

A student reacted lead nitrate solution with sodium bromide solution. A precipitate of lead bromide was produced.

The student measured the mass of the lead bromide precipitate produced.

Figure 6 shows the equipment used.

Figure 6

**0 8****1**

Describe a method to find the mass of the lead bromide precipitate produced.

[4 marks]



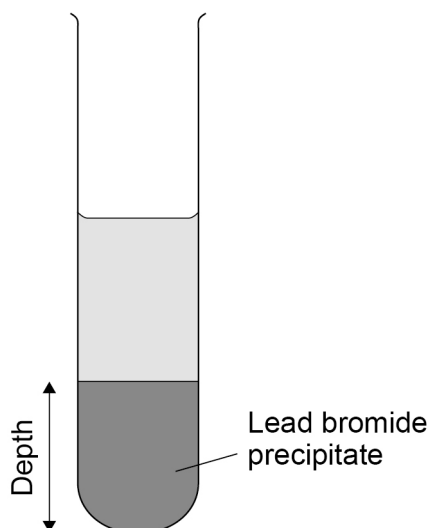
The student investigated the depth of precipitate produced by reacting lead nitrate solution and sodium bromide solution.

This is the method used.

- 1 Add 10 cm³ of lead nitrate solution to a boiling tube.
- 2 Add 2 cm³ of sodium bromide solution to the lead nitrate solution.
- 3 Leave the boiling tube and contents to stand for 10 minutes.
- 4 Measure the depth of the precipitate formed with a ruler.
- 5 Repeat steps 1 to 4 using different volumes of sodium bromide solution.

Figure 7 shows the boiling tube and contents after standing for 10 minutes.

Figure 7



0 8 . 2

Suggest why the student left the boiling tube and contents to stand for 10 minutes.

[1 mark]

Question 8 continues on the next page

Turn over ►



Table 6 shows the results.

Table 6

Volume of sodium bromide solution added in cm ³	Depth of precipitate in mm
2	4
4	8
6	12
8	16
10	20
12	21
14	21

0 8 . 3 Complete **Figure 8**.

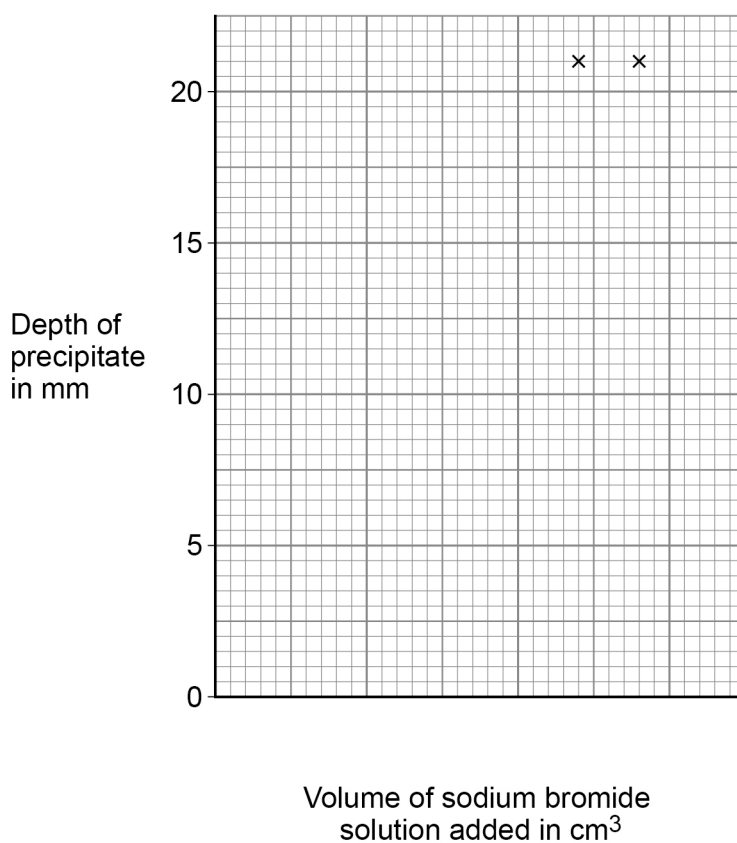
You should:

- use a suitable scale for the x-axis
- plot the data from **Table 6** on **Figure 8**
- draw a line of best fit through the first 5 points.

The last 2 points from **Table 6** have been plotted for you.

[4 marks]



Figure 8

0 8 . 4

Give the reason why there is a maximum depth of precipitate formed in this investigation.

[1 mark]

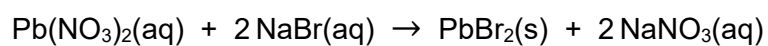
Question 8 continues on the next page

Turn over ►



0	8	.	5
---	---	---	---

The equation for the reaction between lead nitrate and sodium bromide is:



10 cm³ of 0.50 mol/dm³ sodium bromide solution was added to a solution of lead nitrate.

Calculate the maximum mass of lead bromide produced.

Relative formula mass (M_r): $\text{PbBr}_2 = 367$

[6 marks]

Maximum mass of lead bromide = _____ g

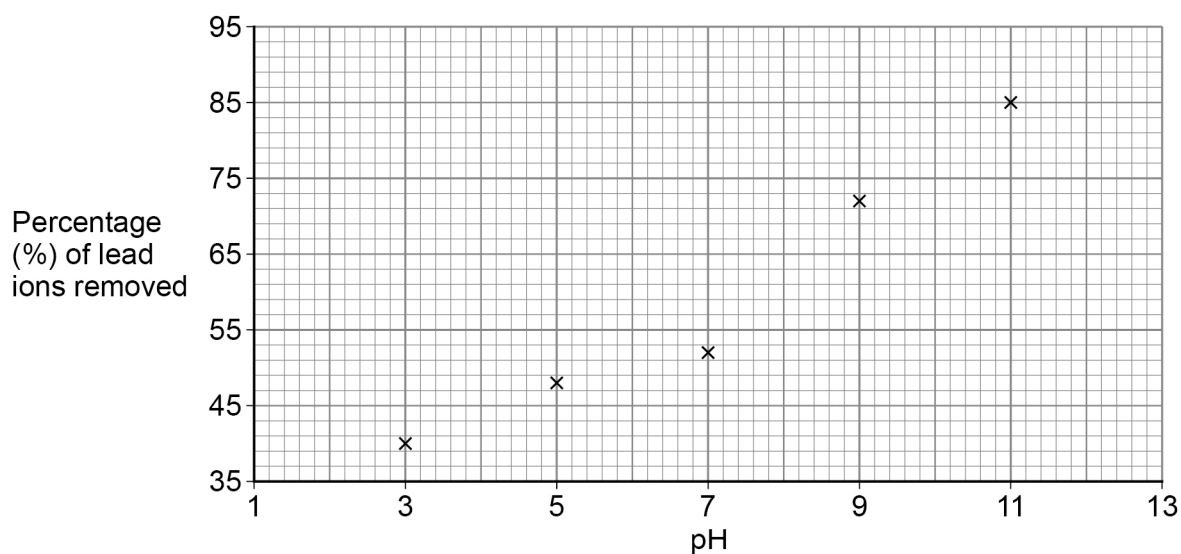


Lead ions can be removed from drinking water by a precipitation reaction producing insoluble lead hydroxide.

0 8 . 6 Scientists investigated the effect of pH on this precipitation reaction.

Figure 9 shows the results of the investigation.

Figure 9



Calcium hydroxide solution can be used to remove lead ions from water.

Explain why calcium hydroxide solution is an effective chemical to remove lead ions from water.

[2 marks]

0 8 . 7 Suggest **one** reason why lead ions are removed from drinking water.

[1 mark]



0 9

This question is about reversible reactions.

0 9 . 1

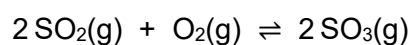
Explain what is meant by 'equilibrium'.

[2 marks]

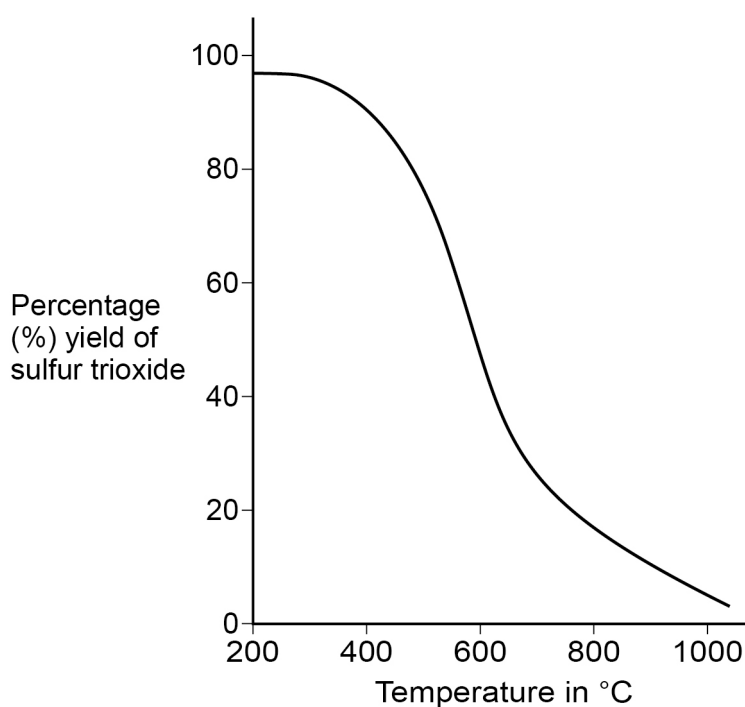
0 9 . 2

Sulfur trioxide is produced during the contact process.

The equation for the reaction is:



The forward reaction is exothermic.

Figure 10 shows the effect of temperature on the percentage yield of sulfur trioxide.**Figure 10**

- vanadium(V) oxide catalyst
- 450 °C
- atmospheric pressure.

You should refer to both yield and rate of reaction.

[6 marks]

[illegible]

8

END OF QUESTIONS



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